

Date : _____

Subject : _____

data = [5, 12, 19, 23, 31, 37, 45, 52, 68, 74, 89, 95]

target = 52, low = 0, high = 11, data[low] = 5, data[high] = 95

$$\begin{aligned} \text{pos} &= \text{low} + \left(\frac{(\text{target} - \text{data}[\text{low}])}{(\text{data}[\text{high}] - \text{data}[\text{low}])} \right) * (\text{high} - \text{low}) \\ &= 0 + \left(\frac{((52 - 5) / (95 - 5)) * (11 - 0)}{1} \right) \\ &= 0 + ((47 / 90) * 11) \\ &= 0 + (0,5222... * 11) \\ &= 0 + (0,57444...) \end{aligned}$$

pos = 5

data[5] = 37, target(52) > data(5), maka ~~low~~ low = pos + 1
= 5 + 1 = 6.

target = 52, low = 6, high = 11, data[low] = 45, data[high] = 95

$$\begin{aligned} \text{pos} &= 6 + \left(\frac{((52 - 45) / (95 - 45)) * (11 - 6)}{1} \right) \\ &= 6 + ((7 / 50) * 5) \\ &= 6 + (0,14 * 5) \\ &= 6 + (0,7) \\ \text{pos} &= 6 + 0 \\ &= 6 \end{aligned}$$

data[6] = 45, target[52] > ^{data}~~low~~[6], maka low = pos + 1
= 6 + 1 = 7.

target = 52, low = 7, high = 11, data[low] = 52, data[high] = 95

$$\begin{aligned} \text{pos} &= 7 + \left(\frac{((52 - 52) / (95 - 52)) * (11 - 7)}{1} \right) \\ &= 7 + ((0 / 43) * 4) \\ &= 7 + (0 * 4) \\ &= 7 + 0 \end{aligned}$$

pos = 7

data[7] = 52, target[52] = data[7], maka ditemukan posisi 52 di indeks 7.

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